



MedAI-CDSS

A Digital System to Support Doctors' Decisions
Using Artificial Intelligence to Improve Diagnostic Accuracy

TECHNICAL & OPERATIONAL PROPOSAL

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1. Introduction and Project Summary

In many African healthcare systems, doctors face a large number of patients within limited time, shortages of diagnostic equipment, and sometimes insufficient information about a patient's history. This situation increases the likelihood of guess-based diagnosis, especially for diseases with overlapping symptoms (for example malaria, typhoid, and other infections common across Africa).

MedAI-CDSS is a proposed digital system designed to help doctors make diagnostic decisions with greater accuracy. This system **is not designed to replace the doctor** — it functions as a "clinical decision support system" (CDSS) that analyzes a patient's initial information (symptoms, how they feel, appearance, and current condition) and provides recommendations on: (a) the likelihood of possible diseases (differential diagnosis), (b) tests that should be conducted, and (c) initial treatment guidance — all subject to the doctor's final confirmation.

"The goal is not to replace the doctor, but to give the doctor a second, digital pair of eyes capable of analyzing large amounts of information quickly and accurately."

2. Problem Statement

- Many patients describe how they feel in everyday language (not medical terminology), so important information is sometimes lost before it reaches the doctor.
- A low doctor-to-patient ratio forces doctors to rush through diagnosis, increasing the likelihood of errors.
- Many tropical diseases (malaria, typhoid, UTI, viral fevers) present with overlapping symptoms, making diagnosis difficult without additional analytical tools.
- Many tests are ordered on a "trial and error" basis rather than being guided by statistical probability (probability-based testing), which delays treatment and increases costs for the patient.
- The lack of digital patient records makes every visit feel like a "fresh start" without connection to prior information.

3. Project Objectives

3.1 Overall Objective

To build a digital system that uses Artificial Intelligence to collect a patient's initial information and help the doctor make diagnostic and testing decisions with greater accuracy, as opposed to guess-based decisions.

3.2 Specific Objectives

- Design a digital intake form/conversation that collects basic patient information in plain language (Swahili/English).
- Build an AI system that analyzes those symptoms and produces a ranked list of possible diseases (differential diagnosis) along with a confidence score.
- Recommend laboratory/imaging tests that align with that likelihood, rather than recommending all tests indiscriminately.
- Produce a clinical summary written in professional language for the doctor, including the reasoning (explainability) behind each recommendation.
- Store patient history digitally so future visits have full context.

- Ensure the privacy and security of patient health data (data privacy) in accordance with national and international regulations.

4. Users and Stakeholders

Stakeholder	Role in the System
Patient	Provides initial information (symptoms, feelings, appearance) via phone, tablet, or with help from reception staff (kiosk/reception).
Nurse / Reception Assistant	Helps record initial vital signs: temperature, blood pressure, heart rate, and so on.
Doctor	Receives the AI summary and recommendations, confirms or adjusts them, and holds the final decision-making authority.
System Administrator (Admin/IT)	Manages system operations, users, and data security.
Ministry of Health / Hospital	Data owner and overseer of usage policy and regulatory compliance.

5. Source of Data

The accuracy of this system depends directly on the quality and volume of data used to train and validate the AI models. Data will be sourced from the following:

5.1 Internal Data (Partner Hospital)

- Patient records (Electronic Health Records - EHR) from the partner hospital, after de-identification/anonymization.
- Records of symptoms reported by patients, tests conducted, and confirmed diagnostic outcomes — this forms the "ground truth" used to train the system.
- Laboratory test results linked to diagnostic outcomes.

5.2 Public and African Datasets

- Symptom-disease datasets available for research, such as those from WHO, MIMIC-III/IV (for general CDSS design reference), and African datasets on malaria, typhoid, and other infectious diseases.
- Official clinical guidelines from the Ministry of Health and WHO, used as a "rule-based validation layer" to verify AI recommendations.
- Peer-reviewed literature describing the relationship between symptoms and diseases, particularly those relevant to African contexts (malaria, cholera, tuberculosis, etc.).

5.3 Data Collected During Use (Continuous Learning)

- Every interaction between patient, system, and doctor (with consent) will be stored as new data to improve the system's accuracy over time (feedback loop).
- Doctor feedback on the accuracy of AI recommendations ("correct" / "incorrect" / "partially correct") will be used as quality labels to continue training the system.

Important note: All patient data will be used with informed consent, and identifying information (name, phone number, address) will not be used to train the AI model — it will be stored separately and protected with encryption.

6. Methodology

The system will be built following the scientific stages of the Medical AI Development Lifecycle, to ensure accuracy, explainability, and safety.

6.1 Stage One — Data Collection and Cleaning

Data from the sources listed in Section 5 will be collected, de-identified, cleaned (removing errors, incomplete or duplicate entries), and organized into a structured format: symptoms, initial test values, brief history, and the final diagnosis confirmed by the doctor.

6.2 Stage Two — Natural Language Processing (NLP)

Since many patients describe their condition in everyday language (for example: "I've felt dizzy and nauseous since yesterday"), the system will use **Natural Language Processing (NLP)** to convert such descriptions into structured clinical symptoms that the AI model can analyze. This will include support for both Swahili and English.

6.3 Stage Three — Disease Probability Prediction Model

The system will use a combination of machine learning techniques to increase both accuracy and transparency:

- **Gradient Boosting Models** (such as XGBoost/LightGBM) — for structured data (vital signs, age, gender, categorized symptoms) — chosen for their strong accuracy in clinical classification problems.
- **Transformer-based Language Models** (fine-tuned on clinical data) — for analyzing free-text patient descriptions.
- **Bayesian Probability Layer** — a statistical layer that uses local epidemiological probability (e.g. accounting for malaria season in a given region) to increase or decrease the weight of a prediction.
- **Rule-based Clinical Guardrails** — fixed rules derived from official WHO/Ministry of Health guidelines, ensuring the system cannot produce a recommendation that contradicts established clinical protocols (e.g. emergency symptoms must always trigger an "urgent referral" recommendation).

This hybrid AI approach ensures the system does not make decisions purely by "guessing from statistics", but combines statistical evidence with verified clinical rules.

6.4 Stage Four — Explainable AI

Every system recommendation will be accompanied by a short explanation of "why" (using techniques such as **SHAP/Feature Importance**) — showing which symptoms contributed most to that particular recommendation. This helps the doctor understand the AI's reasoning rather than receiving it as a "black box".

6.5 Stage Five — Clinical Validation

- The model will be evaluated against a held-out test dataset not used during training, to measure real-world accuracy.
- A clinical advisory board of doctors will review the system's outputs during a pilot phase before full deployment.
- Quality metrics to be used include: *Sensitivity*, *Specificity*, *Accuracy*, *F1-Score*, and *Area Under the ROC Curve (AUC)* — internationally accepted metrics for evaluating clinical diagnostic systems.

6.6 Stage Six — Continuous Learning Loop

Once in use, the system will continue to be improved using real feedback from doctors and test outcomes, under human supervision (human-in-the-loop), to prevent the system from "learning mistakes" without oversight.

7. System Architecture

The system is divided into five core modules that work together:

Module	Function
1. Patient Intake Module	A digital form/chatbot that collects the patient's symptoms, feelings, appearance, and current condition in plain language.
2. Vitals & Pre-screening Module	Nurse/reception staff record initial vital signs (temperature, blood pressure, heart rate, weight).
3. AI Diagnostic Engine	The core of the system — analyzes all information and produces a ranked list of possible diseases with confidence scores and recommended tests.
4. Clinical Dashboard (Doctor View)	The dashboard viewed by the doctor — patient summary, AI recommendations, reasoning behind them, and a space to confirm or adjust the decision.
5. Records & Analytics Module	Stores patient history and provides disease trend data for hospital management and public health planning.

7.1 Information Flow (High-Level Flow)

Patient enters information → System (NLP) converts it into structured symptoms → Initial vitals are added → AI Diagnostic Engine analyzes and produces a ranked list of possible diseases with recommended tests → Doctor reviews the recommendations along with their reasoning → Doctor confirms/adjusts the decision → The final decision is stored as new data to improve the system.

7.2 Recommended Technology Stack

Component	Recommended Technology
User Interface (Frontend)	A web app and mobile app for patients, nurses, and doctors.
Backend	API services connecting all modules and communicating with the AI Engine.
AI/ML Engine	Machine learning and language models trained on the clinical data described in Sections 5 and 6.
Data Storage	A secure, encrypted database compliant with international health data privacy standards.
Security	User authentication, role-based access control, and audit logs.

8. Privacy, Security, and Clinical Ethics

- **Informed Consent:** Every patient will be clearly informed that the system uses AI as an assistive tool, and has the right to decline the use of their data for research/system improvement purposes.
- **De-identification:** Information that identifies a patient will not be used to train the AI model; it is separated from clinical data.
- **Human-in-the-loop:** The system does not make the final decision — the doctor retains full authority over diagnosis and treatment at all times.
- **Regulatory Compliance:** The system will be built in accordance with national data protection laws and international guidance on AI ethics in health (WHO Ethics & Governance of AI for Health).
- **Encryption and Auditing:** All data will be stored with encryption at rest and in transit, and every access to data will be logged for audit purposes.

9. Success Metrics

Metric	Description
Diagnostic Accuracy	Percentage of AI recommendations that match the final diagnosis confirmed by the doctor.
Sensitivity / Specificity	The system's ability to correctly identify existing diseases (sensitivity) without producing too many false alerts (specificity).
Time-to-Decision	Reduction in the time needed to reach an initial testing decision, compared to the current approach.
Doctor Acceptance Rate	Percentage of doctors who find the system's recommendations useful in their daily practice.
Reduction in Unnecessary Tests	Reduction in patient costs resulting from probability-guided testing rather than "testing everything".

10. Anticipated Challenges and Mitigation Strategies

Challenge	Mitigation Strategy
Limited high-quality internal data	Begin with public international/African datasets, then gradually add internal data through the pilot phase.
Doctor resistance/concern that "AI will replace them"	Clear training and communication that the system is an "assistant", not a replacement; involve doctors from the design phase onward.
Network/power outages in rural areas	Design the system to work in an offline-first mode and sync data once connectivity returns.
Algorithmic bias	Ensure training data represents diverse patient groups (age, gender, region) and evaluate the system's performance across those groups.
Health data privacy	Encryption, de-identification, and strict data access policies as described in Section 8.

11. Implementation Roadmap

Phase	Key Activities	Estimated Duration
Phase 1	Requirements gathering, data access, and agreement with the partner hospital.	Weeks 4 - 6
Phase 2	Data cleaning and development of the initial AI model (proof of concept).	Weeks 6 - 10
Phase 3	Building the applications (Patient Intake, Doctor Dashboard) and integrating with the AI Engine.	Weeks 8 - 12
Phase 4	Internal testing and clinical validation with the advisory board of doctors.	Weeks 4 - 6
Phase 5	Pilot testing at the hospital with a small number of patients, under close supervision.	Weeks 8 - 12
Phase 6	Refinement based on feedback, followed by official launch and a continuous improvement cycle.	Ongoing

12. Required Team

- **Data Scientist / ML Engineer:** Builds and improves the AI models.
- **Software Developers:** Build the web/mobile applications and backend system.
- **Clinical Advisor (Doctor):** Validates the clinical accuracy of the system at every stage.
- **Data Security/Privacy Officer:** Ensures compliance with privacy regulations.
- **Project Manager:** Coordinates all activities and communication with the partner hospital.

13. Conclusion

MedAI-CDSS aims to bridge the significant gap between the large number of patients and the limited time available to doctors, by using AI as an intelligent assistant that increases diagnostic accuracy without removing decision-making authority from the doctor. By combining internal data, public international/African datasets, and modern machine learning techniques together with solid clinical rules, this system has the potential to reduce guess-based diagnosis, reduce unnecessary testing, and ultimately improve health outcomes for the community.

The next step is to reach agreement with a partner hospital on data access, form the initial team, and begin Phase 1 of implementation.

For further questions or an in-depth discussion of this proposal, we are ready to expand any section with additional technical or financial detail as needed.